Assignment 2

September 21, 2025

The Mission: Escape Through the City

You wake up in an unfamiliar city. The streets stretch like a maze of long blocks and hidden alleys. Somewhere in this city lies your **destination**—a safe house you must reach before time runs out.

But the city is full of dangers:

- **Obstacles:** Some roads are completely blocked. If you bump into one, you get hurt.
- **Rainy Conditions:** The slippery ground makes your moves unreliable. Even if you plan to go north, you might slip sideways instead.

Every step costs you valuable time, so you must choose your path wisely. Your objective is simple:

> **Find the safest and fastest route to the destination, avoiding dangers and handling the uncertainty of rain, all within the strict time constraint.**

Mapping to the GridWorld

We model this city as a **GridWorld environment**:

- Each cell of the grid is a **state**.
- **Obstacles:** If the agent attempts to move into a blocked cell, it stays in place but receives a penalty of -10 (to depict getting hurt).
- **Goal:** A terminal state, but with no additional positive reward.
- **Actions:** Up, Down, Left, Right.
- **Stochasticity (Rain):** Intended action may fail, and the agent slips into a neighboring cell instead.
- **Rewards:** Each step costs time and is penalized with -1; hitting an obstacle gives -10; reaching the goal ends the episode with no extra reward.

Step by Step Mission

We will tackle the problem in stages, gradually increasing the difficulty. This way, you can first build confidence with Dynamic Programming methods before handling the full city challenge.

1. **Stage 1: Sunny Day, Clear Roads** No obstacles, no slipping. Verify that Value Iteration and Policy Iteration correctly find the shortest path.
2. **Stage 2: Obstacles Appear** Some roads are blocked, and bumping into them hurts. Your algorithms must now avoid obstacles while minimizing travel time.
3. **Stage 3: Rainy Day Challenge** The full scenario: obstacles remain, but slippery roads make actions uncertain. Here your methods must handle both danger and stochasticity.

Through these stages, you will see how Dynamic Programming methods evolve from solving simple cases to mastering complex, uncertain environments.

Provided Code Base

To help you get started, we provide a nearly complete code base that contains the full environment setup along with placeholders where you will implement the core algorithms. You need to run '**main.py**' for running experiments.

What You Will Receive

- A zipped folder containing the GridWorld environment and utility scripts.
- Implementations of the overall framework, with key parts marked as `TODO` in '**agents.py**' file:
  - Value Iteration
  - Policy Evaluation
  - Policy Improvement
- An **oracle** that ensures each student works with a unique configuration (different start, goal, and obstacle placements).

Requirements

- Python version 3.7 – 3.13 (compiled oracles are provided for these versions).
- Packages: `numpy`, `matplotlib` (install via `pip` if not already available).

Note

- If your Python version is outside the supported range, or if you encounter issues running the code base, please contact the TAs.

Your Mission Tasks

In this assignment, you will gradually increase the complexity of the city environment and analyze how Dynamic Programming methods behave under different conditions. At each stage, you will run experiments, report observations, and connect your findings back to the mission.

Stage 1: Sunny Day, Clear Roads

- Run the environment without obstacles and without slipping.
- Compare **Value Iteration** and **Policy Iteration** in terms of convergence speed.
- Experiment with at least 5 different discount factors γ (e.g., $0.9, 0.8, 0.7$ etc.) for policy and value iterations (You can change them jointly or independently).
- Report whether the choice of γ affects the speed of convergence or not.

Stage 2: Obstacles Appear

- Add blocked roads (obstacles) to the environment.
- Repeat the same experimentation as done in Stage 1 along with convergence analysis.
- Compare the results with the obstacle-free case.
- Reflect on how obstacles change agent choices, just as avoiding dangerous alleys in the city while still trying to be fast.

Stage 3: Rainy Day Challenge

- Now add both obstacles and rain (slippery moves).
- Test with three slip probabilities: 0.0001, 0.001, and 0.01.
- Observe how the algorithms behave:
  - Does one converge while the other gets stuck?
  - Which one shows faster convergence?
- You may adjust parameters of the algorithms (you need to find appropriate ones) to attempt convergence. Report your findings even if the attempts fail.

Note

At each stage, a successfully learned policy will be saved as an image. You must include these images in your report, along with your analysis and findings. This will clearly demonstrate how the algorithms adapt as the city environment becomes progressively more challenging.

Deliverables

You are required to submit **a single PDF document** containing the following:

- Implemented code snippets for the functions marked as `TODO` in the provided code base.
- Results and analysis for all tasks described in the previous section, including convergence behavior, parameter experiments, and observations.
- Images of the **Learned Policies** obtained at each stage.

All components must be compiled into a single PDF file and submitted as your final report.

Important Instructions

- This is an **individual assignment**. Collaboration or code sharing is strictly prohibited.
- Deadline: 12$^{\textbf{th}}$ **October 2025, 11:59 PM**. Late submissions will not be accepted.
- The submitted code and reported results must be consistent. If the code and results do not align (e.g., correct code with incorrect results, or correct results without supporting code), no marks will be awarded for those parts.

Good luck, and may your policies guide you safely through the city!